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Big Loans to Small Businesses: Predicting Winners and Losers in an Entrepreneurial Lending Experiment†

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We experimentally study the impact of relatively large enterprise loans in Egypt. Larger loans generate small average impacts, but machine learning using psychometric data reveals "top performers" (those with the highest predicted treatment effects) substantially increase profits, while profits drop for poor performers. The large differences imply that lender credit allocation decisions matter for aggregate income, yet we find existing practice leads to substantial misallocation. We argue that some entrepreneurs are overoptimistic and squander the opportunities presented by larger loans by taking on too much risk, and show the promise of allocations based on entrepreneurial type relative to firm characteristics. (*JEL* C45, D22, G21, G32, L25, L26, O16)

When credit markets are characterized by asymmetric information, relational contracts dominate, and a specific lender can become the de facto monopoly provider of credit to a set of firms wishing to access larger loans (Ghosh and Ray 2016). In this circumstance, an individual lender's credit allocation choices can become important determinants of the efficiency of capital allocation and hence aggregate productivity (Hsieh and Klenow 2009). Furthermore, if lender credit allocation is inefficient, then the average large loan will generate more default and less firm profit than it could, and equilibrium would be characterized by both misallocation and smaller-than-optimal loan sizes.

Despite the potential importance, little is known about the quality of lender decisions beyond that which can be inferred from aggregate exercises such as

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those presented in Hsieh and Klenow (2009, 2014).[1] This dearth of knowledge likely stems from three difficulties an analyst faces when attempting to understand the quality of lender credit allocations. First, if poor allocation of credit means large loans have high default rates or are unprofitable for borrowers, equilibrium loans may be too small to reveal heterogeneity. Second, to understand the quality of allocation decisions, one must collect appropriate data to estimate heterogeneous treatment effects and use methods that avoid overfitting. Finally, there needs to be a credible identification strategy, one that generates both a sufficient quantity of loans to firms that would not normally be allocated larger amounts of credit and determines which firms would have received loans in a business-as-usual loan size expansion.

We collaborated with the Alexandria Business Association (ABA), a large Egyptian lender, to design and implement an experiment that would overcome the above difficulties. ABA's senior management understood that there was demand for larger loans but was concerned about risk, both for themselves and their clients. Our collaboration was designed to understand whether larger loans could be beneficial and whether credit allocation decisions could help to mitigate risk.

ABA first worked to select a sample of firms that it believed might benefit from a larger loan. They then conducted a randomized experiment within this sample, in which treatment borrowers were offered loans four times larger than their previous loans and control borrowers were offered loans that were twice as large.[2] Borrowers were given some flexibility regarding loan size and duration within these limits, a design that we believe captures the impact of increased credit access. The experiment meets all the challenges discussed above. Large loans were offered even to firms that would not normally qualify for loans of this size, extending well beyond normal lending practice for ABA. To measure heterogeneous treatment effects, we collected novel psychometric data at baseline. These data were designed to capture characteristics of the owner relevant to entrepreneurial success. We combine these data with modern machine learning methods from Chernozhukov et al. (2023) to identify group average treatment effects for four quartiles. We call those in the top quartile of likely entrepreneurial success "top performers" and those in the bottom quartile "poor performers."

Having identified top and poor performers, we use two strategies to understand whether larger loans would be allocated to top performers in a business-as-usual expansion. First, we believe a reasonable theory of practice is that lenders target capital expansion toward firms that demonstrate success with small loans, believing

[1] Recent work has focused on a related question, whether loan officers can help choose firms that are likelier to repay and more profitable for the bank, as well as whether loan officer recommendations reveal bias (e.g., Dobbie et al. 2021; Fisman, Paravisini, and Vig 2017; and Cole, Kanz, and Klapper 2015). Particularly closely related to our motivating questions is Rigol and Roth (2021), which shows that contractual incentives for loan officers can easily be misaligned, even against lender profits. Specifically, they show that loan officers incentivized to reduce average default in their client set do not recommend their lowest default clients to receive larger "graduation loans," which are provided by a separate department of the same lender. None of these papers considers the profitability of the borrowing firm, which is our focus, and what matters for aggregate productivity.

[2] Standard operations provided borrowers the ability to borrow at 1.5 times their previous loan. The decision to provide the control group with 2× loans was made by the bank at the suggestion of loan officers because they believed this would make the randomization seem fairer. This choice has some advantages; for instance, it keeps control borrowers engaged with the lender. On the other hand, it limits our ability to speak about how the larger loans compare to status quo lender policy. Borrowers generally had flexibility in determining their loan term, and many borrowers who received the larger loan opted to extend the time for repayment, which is an endogenous response to the increased access to credit.

these entrepreneurs to be of a type that will be consistently successful. This theory suggests preferential access to large loans would be given to firms that see the largest change in profits between baseline and endline when assigned to the control group. We examine whether top performers are more or less likely to fall into this category. Second, like many lenders, ABA relies heavily on loan officer opinions to determine credit allocation, and loan officers are strongly incentivized to avoid default. We asked a subset of loan officers at baseline to assess whether individual firms would be more likely to default if given a larger loan, and we look at whether loan officers perceive top performers as more or less likely to increase their risk of default.

Our results reveal the potential importance of credit allocation, even *within* the set of ABA borrowers. We find mostly null average impacts from the larger loans but that there is important heterogeneity in treatment effects, along with strong evidence of misallocation. Entrepreneurs in the top quartile of treatment effects for profits, i.e., the top performers, increase profits by 55 percent (SE = 21 percent) of the control group mean. Poor performers, on the other hand, are harmed, experiencing a 52 percent (SE = 23 percent) profit reduction. The same pattern holds for related and downstream outcomes. Top performers increase their wage bill (140 percent, SE = 29 percent), productivity (6.2 percent, SE = 1.0 percent), and household expenditures (46 percent, SE = 10 percent). Poor performers, on the other hand, lose 1.7 (SE = 0.55) employees and greatly reduce their wage bill.

To gauge the importance of these magnitudes, we compare an intentional lending strategy that targets likely high performers rather than one that lends to the full set of firms identified as viable by the lender for larger loans. Targeted lending would improve the average treatment effect for productivity from zero to about one-half of a standard deviation and would increase the treatment effect on monthly profits by about 47 percentage points. Across 1,000 firms, this would amount to an increase in aggregate monthly profit of about £E7 million (approximately US$400,000).

However, the top performers are among the firms least likely to be offered large loans. First, relative to poor performers, top performers see a smaller increase (or larger decrease) in profits between baseline and endline when they are placed in control, and this difference is statistically significant. This suggests that loan officers would be unlikely to see top performers as successful firms and hence unlikely to recommend them for further loan expansion.

Second, directly measured loan officer beliefs strongly militate against lending to top performers. Loan officers believe that large loans will increase the chance of default by 28 percent for poor performers but by 46 percent for top performers. As mentioned above, loan officer incentives revolved heavily around default rates rather than total returns or firm profits. At the time of the experiment, the majority of loan officer compensation was an incentive payment, with loan officers receiving no incentive payment if their portfolio's repayment rate was below 97 percent. As a result, loan officers are unlikely to have suggested extending credit to top performers in the absence of our experiment.[3] However, we find that loan officers believe

[3] Despite loan officer beliefs, we find no evidence that top performers are more likely to default. Full loan default is very rare since failure to repay a loan could result in a prison term. Typically, the lender only pursues legal avenues in cases where they suspect fraud (rather than just low business outcomes). For our study, the lender

that top performers would be more likely to increase revenues with a big loan, in line with our empirical results. This suggests that a simple change in loan officer incentives to emphasize firm revenues might lead to better allocation of capital. Indeed, ABA revised its incentive system in response to our findings, moving to a system based on the profitability of the loan officer's portfolio as a whole.

Two points are worth noting about the heterogeneity results. First, using only "standard" data of the type that might be routinely collected by a lender (e.g., business performance and demographics), we do not discover any statistically significant heterogeneity.[4] The heterogeneous treatment effects reported above are identified using a combination of psychometric and cognitive data.[5] This suggests both that a lender seeking to better allocate larger loans will need to collect more data than usual and that the characteristics of the *entrepreneur*, rather than the characteristics of the *firm*, are important for determining the impact of increased access to credit for small firms. While important details need to be worked out—including, for example, understanding the extent to which psychometric testing is manipulable—our results suggest the *possibility* of improving allocations through the use of better data collection.

Second, while poor performers perform *relatively* poorly with large loans, they actually have higher *relative* profits in cross-sectional analysis among the control group—those getting smaller loans. At first glance, this reversal of fortune may seem counterintuitive; it is contrary to the usual model in which entrepreneurs are credit constrained and high-ability entrepreneurs squeeze relatively more out of scarce resources, whether at low or higher levels of capital. However, we believe that the reversal of fortune is a straightforward implication of two hypotheses that are reasonably uncontroversial. First, a large literature in psychology suggests that there is substantial heterogeneity in individuals' optimism levels, with a sizable proportion of the population being unrealistically optimistic (Carver, Scheier, and Segerstrom 2010; Peterson 2000; Weinstein and Klein 1996).[6] Second, while the finance literature widely accepts that there is a risk-reward trade-off, it is unlikely that expected gains can be increased indefinitely by taking greater and greater risk; i.e., the expected income gains that come from taking greater risk ought to be subject to diminishing returns.

We provide a simple model in Section VI that shows how these two assumptions can combine to explain our results. The argument is straightforward. We think of our poor performers as being overoptimists and our top performers as being realists. A small loan opens up the opportunity for taking a risky investment, and overoptimists take more risk than the realists, leading to a larger expected reward. A larger loan, however, opens up even more risk-taking opportunities and induces

agreed to not pursue prison terms for anyone in the sample. Borrowers do, however, often deviate from the precise repayment schedule, and it can be costly for ABA to engage in additional monitoring and enforcement activities. Imprisonment is not an automatic consequence of default; it requires the lender to initiate legal proceedings and for them to be successful in court.

[4] For a more detailed description of both the standard and psychometric data, see Section IC.

[5] Cognitive data include measures such as Raven's matrices and also risk aversion measures.

[6] We use the term "optimism" throughout the paper, but several other closely related concepts could also explain our results. For example, overconfidence, which is optimism about one's own performance, and being risk loving could also lead an agent to take on too much risk. We discuss this issue in more detail in Section VI, but we think of optimism as a composite of this set of related attributes.

the overoptimists to increase their risk taking to the point where it actually leads to a reduction in expected reward. No such effect occurs for the less optimistic, who are better able to appreciate that very high risks are not accompanied by strong returns. These dynamics are in line with previous work that has shown a relationship between entrepreneur optimism and business outcomes that is sometimes negative (Frese and Gielnik 2014; Hmieleski and Baron 2009) and sometimes positive (Hilary et al. 2016).

We are not able to test the assumption of decreasing returns to risk taking, but we can use our psychometric data to shed light on the hypothesis that the poor performers are overoptimistic. We do two things. First, we use common machine learning methods to determine which psychometric variables are most predictive of being a poor performer. This process highlights seven psychometric questions that are related to optimism; poor performers respond more positively to statements like "I can think of several solutions to any problem," suggesting optimism about risk taking. Poor performers are also more likely to agree with statements indicative of a gung ho attitude, such as "I tend to act first and worry about consequences later" and "When I make decisions I usually go with my first, gut feeling." Second, we look manually through our set of 50 psychometric questions to isolate those that we think are measuring optimism and test if they are predictive of being a poor performer. We find that poor performers are more likely to agree with statements like "When I make a business decision it is almost always the right decision," "I have always believed I am going to be successful," and "I prefer to focus on opportunities rather than risks." Overall, while not dispositive, we believe that these correlations give credence to our optimism-based interpretation.

Our paper joins a growing literature that uses different indicators to document heterogeneous firm-level returns to capital. Hussam, Rigol, and Roth (2022) finds that individuals are able to identify which peers have the highest returns to capital. Beaman et al. (2020) finds that farmers who borrow have higher returns to capital than those who do not. Banerjee et al. (2019) shows how experienced entrepreneurs have much higher returns to credit, while Meager (2020) shows how the returns to credit can be very positive for some but also negative for others. Crépon, Komi, and Osman (2024) shows that there is substantial heterogeneity in the returns to both cash drops and loans and that this may help to explain the difference in impacts between cash drop experiments in the spirit of De Mel, McKenzie, and Woodruff (2009) and the more recent microcredit experiments. Also related is McKenzie and Sansone (2019), which finds that neither experts nor machine learning methods predict future entrepreneurial success using "standard" loan application and business data. This question differs from ours in that they are predicting future enterprise success, not treatment effects, and they have more limited psychometric data.

We also contribute to the literature on randomized evaluations of credit interventions. The majority of studies in this literature look at the impact of expanding access to micro-size loans and find modest effects (see Banerjee, Karlan, and Zinman 2015; Meager 2019). In contrast to these studies, we study a different margin of credit expansion—access to much larger loans than firms are normally able to access—and we concentrate on the importance of heterogeneity rather than average effects. Our work is also closely related to papers that look at alternative margins on which loan access can be altered in developing countries. In recent important work, Breza and Kinnan (2021) studies the impact of the canceling of all microcredit access in Andra Pradesh due to the Krishna loan crisis. They find large impacts on household consumption. The margin on which this experiment alters access is, like our study, very different from the norm: canceling borrowing for all rather than marginal borrowers, removing credit rather than adding credit, and general equilibrium rather than partial. Also related is Banerjee and Duflo (2014), which studies an Indian program that expanded preferential credit access to larger firms (with more than US$150,000 in capital stock). This expansion led to substantial investment and increased profits. Finally, Bari et al. (2021) examines the provision of in-kind loans to existing businesses, which, like our treatment, are four times the size of previous loans. They find statistically significant and positive average effects but do not concentrate on the allocation decision on which we focus here.

I. Study Setting and Sample Characteristics

A. *Implementing Partner*

We partnered with Alexandria Business Association, a nonprofit organization and the largest microfinance institution (MFI) in the Middle East by number of clients. At the time of the study, ABA had over 400,000 borrowers, and fewer than 1 percent of those borrowers had loans over US$1,000\£E13,000.[7] Senior management expressed interest in expanding the size of loans offered to small and medium enterprises (SMEs). Immediately prior to our project, ABA policy allowed for new loans to increase by up to 50 percent after successful repayment of a prior loan. Loan size expansion ceases when the firm stops demanding more credit, the loan officer deems the loan to be too risky, or they reach £E150,000, which was the regulatory maximum for MFIs in Egypt at the time of the study.[8]

At the time of our study, ABA provided both individual and group loans, and our intervention focused on the individual lending side of the portfolio. Individual loans began at US$115\£E1,500 and had a ceiling of £E100,000. Annual interest rates varied slightly depending on market conditions but ranged from 14 percent to 17 percent.[9] Borrowers could take loans of various maturities, but the majority of loans were provided on 12-month terms. Borrowers could go to a branch office to apply for a loan, but many loans were initiated directly with loan officers who often visit businesses to recruit new borrowers. The application process required individuals to commit to repayment and to include a guarantor. Lending criteria were focused on an individual's ability to prove that they have a functioning business and a sound plan for the use of the funds, as well as the loan officer's subjective opinion of whether the borrower would be able to pay back the loan.[10] ABA provided both

[7] The average exchange rate during the disbursement of the loans was approximately £E13 per US$1. The PPP conversion factor was about 2.9 during the study period. We use the exchange rate for all conversions unless stated otherwise.

[8] Borrowers could request a larger than 50 percent increase, but that required approval from the deputy CEO and COO and only happened in extremely rare cases.

[9] In this context many borrowers may be concerned about the Islamic prohibition on paying interest, but a ruling by Egypt's main religious authority, Al Azhar, states that interest rates charged by nonprofit lenders are not prohibited.

[10] The guarantor is a person who would be legally liable for the debt in the case of default.

household and enterprise loans, but loans over £E20,000 were provided only to registered businesses. This requirement was relaxed for our study, allowing many unregistered enterprises to get loans larger than £E20,000.

In Egypt, as in many developing countries, there is a "missing middle" of credit availability that may contribute to the lackluster growth of small firms: MFIs typically offer loans up to a maximum of about £E5,000 (approximately US$280), whereas banks typically start loan sizes at £E50,000 (approximately US$2,800) and have formality requirements that small enterprises often cannot meet. We conjectured that this gap in the lending market could, in part, be attributed to the combination of asymmetric information, which leads to relationship-based lending starting from when firms are micro-sized, and poor credit allocation decisions reducing the profitability of larger loans. Rigol and Roth (2021) provides a related explanation.

B. *Experimental Design*

The loans were implemented under a program called *Tamouh*, which translates to "ambition." To form an eligible cohort of borrowers, the bank first informed loan officers about the lending program and asked them to suggest clients who would be appropriate for a much larger loan. Loan officers then communicated the details of the program to clients they thought would be a good match, telling them that if their application was approved, they would be randomly allocated to either a 4$\times$ or 2$\times$ larger loan. Loan officers were selective and only invited clients they thought would be a good match. Most suggested only a small proportion of their set of existing borrowers. Existing clients who were interested and deemed eligible then applied for a loan at least four times the size of their previous loan.[11] It is important to note that this design implies the control group was a small deviation from prior policy, in which loan sizes increased by at most 50 percent, except in rare cases. The decision to provide the control group with 2$\times$ loans rather than the existing policy of 1.5$\times$ loans was made by the bank at the suggestion of loan officers, who believed that this would make the randomization seem fairer. Since the selection into the experiment is endogenous, we cannot cleanly interpret a comparison of the 1.5$\times$ existing policy to the 2$\times$ in the control group.

Part of the capital for the *Tamouh* loans was provided by Silatech, a regional philanthropic foundation, which agreed to cover some of the risk of default. The partial guarantee was negotiated because of ABA's concerns that the larger loans would lead to default. This fear of default is common among the small and medium enterprise lenders we have interacted with, and part of the purpose of this project was to understand whether this fear is well founded and, if so, whether better credit allocation choices could alleviate risk. The existence of the guarantee was known to the Board of Directors of ABA but was intentionally kept from loan officers to avoid any change in enforcement efforts. At the time of the experiment, the majority of loan officer compensation was an incentive payment that depended on the number of loans, not the size of loans, and also included a repayment condition. If a loan officer's portfolio repayment rate was below 97 percent, they did not receive any

[11] Although the plan had been to allow treatment loans to be up to ten times the size of the previous loan, in practice, ABA let treatment loans be typically four times larger but not more.

incentive payment; for repayment rates above 97 percent, but at or below 98.5 percent, they received 50 percent of their incentive payment; and for repayment rates above 98.5 percent, they received their full incentive payment. Because repayment rates were based on default relative to overall portfolio size, loan officers expressed concern that the *Tamouh* loans could harm their overall compensation. To appease this concern, senior management added an additional incentive for loan officers, which provided a fixed payment for each client who ended up in the experiment. The fact that ABA felt the need to make this payment reveals the importance of loan officer beliefs in determining credit allocations. Ultimately, all of the loans in the experiment, both treatment and control, were repaid and thus had no adverse effect on the loan officer's portfolio repayment rates.

To apply, invited borrowers fill out a standard loan application as well as an in-person baseline research survey, which was conducted by ABA employees from the nonfinancial services division.[12] ABA surveyors informed borrowers about the goals of the survey and that their data would neither be shared with ABA's financial arm nor affect ABA's lending decision in any way. The baseline survey included detailed questions about the borrower's businesses, demographic questions, and nonstandard types of data, such as cognitive and psychometric questions. Afterward, a central credit committee reviewed the borrower's application (without access to the baseline survey answers) and made a lending decision. In all cases, the credit committee required that borrowers had successfully repaid at least three prior loans. Approved loans were grouped into batches for randomization. For the randomization, we worked closely with ABA to randomize 31 batches of applicants over 13 months. For each batch, ABA would send the research team a list of between 1 and 133 applicants. For each applicant, we were given the full application material but only used the identity of the loan officer for the purpose of stratification. To stratify, we employed a simple block randomization routine by loan officer. When there was an odd number of applicants with an individual loan officer, we assigned the additional applicant with an independent 50 percent probability to treatment or control. Our final sample consists of 1,004 borrowers, nominated by 168 loan officers. Disbursement of the loans for the sample took place in 2016 and 2017.

Aside from the loan size, the loans adhered to ABA's normal loan terms, monthly repayments, and monitoring and enforcement protocols (a more detailed description of these is included below, under description of the first-stage results). ABA's loan policies allow borrowers some discretion on the period over which to repay the loan. Borrowers could choose a loan term between 6 and 24 months. We will find below that the treatment group responded to the larger loans by requesting longer loan terms on average. Hence, when comparing our intervention to others in the literature, our treatment group gets more capital and longer to repay it.

[12] The data collection had to be implemented by employees of the organization due to data collection regulations in Egypt at the time of the study. They were part of ABA's nonfinancial services division and accompanied by trained supervisors from Silatech.

C. *Data Collection*

We use two data sources—in-person surveys and administrative data on loan performance. We implemented three rounds of in-person surveys: a baseline implemented before firms were randomized into treatment and control groups, a first follow-up implemented on average 20 months after disbursement of the loans, and a second follow-up implemented on average 30 months after the disbursement of the loans.[13] The in-person surveys collected two types of data. They collected standard data on borrower characteristics and the business, cognitive data, risk preference data, and "psychometric data" collected through a set of 50 statements meant to characterize borrower personality traits.

Standard Data on Business and Borrower Characteristics.—All three surveys included standard information on business and borrower characteristics. For business questions, we included sector, profits, revenue, expenditures, and the number of employees and their wages. For borrower characteristics, we included age, experience, and education. We also measured household expenditure using a survey question that asked for total expenditure on a set of commonly consumed food items over the past week; a second question asking total expenditure on a set of less regular expenditures, such as water, electricity, and transportation, over the past month; and finally, a question asking total expenditure on irregular purchases, like clothes and mobile phones, over the past year. We use baseline versions of these measures to check for balance across treatment and control and as inputs into the machine learning algorithms. We use endline measures of these variables as our primary outcomes.

Psychometric, *Cognitive Data*, *and Risk Preferences*.—We included 50 "psychometric" questions in our baseline survey. These data take the form of statements that respondents were supposed to assess on a scale from 1 to 5, with a "1" indicating that they disagree strongly with the statement and a "5" indicating that they agree strongly with the statement. The statements are meant to measure which psychological traits are associated with the respondent's personality. These include traits considered to be in the "Big 5" in the psychology literature, such as conscientiousness, extroversion, and agreeableness (Digman 1990). Examples of the statements include "I am good at making last minute changes to plans," which is associated with conscientiousness, and "Before I go to sleep I think 'What could I have done better today?'." We also included statements that were shown to be associated with people's proclivity for entrepreneurship. These statements include "In my group of friends I am the most creative person," and "I have a strong desire to be successful in life" (Ahmetoglu, Leutner, and Chamorro-Premuzic 2011; Ahmetoglu et al. 2017). The 50 statements at baseline were a subset of a 140-statement list that was produced for this study.[14] The remaining statements were included in the follow-up

[13] We intended to survey everyone at approximately the same time, but sometimes it took longer to locate some of the borrowers, and other times we were delayed in renewing the permits necessary to collect data in this context.

[14] These psychometric statements were agreed upon after close consultation with a set of psychologists who have published extensively on entrepreneurship traits (Ahmetoglu, Leutner, and Chamorro-Premuzic 2011; Ahmetoglu et al. 2017; Leutner et al., 2014).

surveys. The statements were split across the three surveys in an ad hoc fashion given survey time constraints and our lack of a clear theory to guide choices.[15]

We measure cognitive ability during the baseline survey using Raven's matrices, digit span recall, and a set of questions testing financial literacy knowledge. We also ask questions at baseline to assess risk aversion, asking both about overall perception of risk taking and a question about a hypothetical risky investment.

A full list of the risk, financial literacy, psychometric, and cognitive questions is included in the Survey Questions section of the online Appendix.

Administrative Data and Loan Officer Perceptions.—Administrative data from the lender provide a detailed account of every payment made by the borrower, including the amount of the payment, the date of the payment, and the proportion of the payment that went to principal, interest, and any associated late fees. These data cover the 5 loans prior to the experimental loan as well as any loans up to 30 months after the experimental period.

We also implemented a loan officer survey. The survey collected demographic information about the loan officers as well as information about the loan officers' expectations regarding business and loan performance for the borrowers in the case where the borrowers were allocated to treatment or to control. We asked loan officers to grade borrowers on their "ability to repay the loan" on a 1–10 scale. We asked them the question twice, once about their perception of repayment ability in the case that the borrower received the larger loan and again in the case that the borrower received the smaller loan. In our analysis, we score these as binary variables that combine the two responses, equal to one if the loan officer perceives the larger loan will be more likely to lead to default than the smaller loan. We have similar data for loan officer predictions as to the effect loans would have on borrower revenues. Due to logistical constraints, we were only able to collect these data before randomization for about a third of the sample of loans (293/1004 borrowers), and we do not use data for loans where the predictions were elicited after the loan had been received.[16] On average, loan officers expect the likelihood of repayment to decrease with the larger loan relative to the smaller loan for 31 percent of borrowers and for 32 percent of borrowers to have a greater ability to increase revenue with the larger loan than with the smaller loan.[17]

Sample Characteristics.—Table 1 reports sample summary statistics from the baseline survey and balance verification for the randomization. Borrowers in the control group are on average 41 years old with 9 years of education; about 20 percent of the sample are women. For the businesses, about a third are registered, the average tenure is 12 years, and the average number of employees is 0.65 (36 percent have any

[15] We included the questions in follow-up surveys with the hope that the psychometric measures would be time invariant and treatment invariant and thus enable us to treat them all as "baseline" measures. To test for this, we included some questions at both baseline and follow-up, but we find enough changes over time to eliminate the ability to use the questions asked at follow-up in the heterogeneity analysis.

[16] An F-test of the hypothesis that the 293 firms that we can include have similar characteristics to the remaining borrowers fails to reject when we include 15 key business and demographic variables.

[17] If the survey was conducted after randomization and loan provision, we find that treatment assignment affects loan officer perceptions; thus, we only use the loan officer surveys that were completed prior to randomization and loan provision.

Table 1—Baseline Balance and Take-Up Composition

	Control mean (1)	Control difference (OLS) (2)	Obsrvs. (3)	Predictors of take-up in control (4)	Predictors of take-up in treatment (5)	p-value for test of (4) = (5) (6)
Baseline value of standard variables						
Age	40.71	−0.49	1,003	0.00	0.00	0.82
	{9.96}	(0.65)		(0.02)	(0.02)	
Female (=1)	0.18	0.03	1,004	0.04	0.04	0.90
	{0.39}	(0.03)		(0.02)	(0.01)	
Years of education	8.71	0.05	996	−0.02	−0.02	0.92
	{4.50}	(0.29)		(0.02)	(0.02)	
Business is registered	0.34	−0.01	1,003	0.04	0.01	0.13
	{0.48}	(0.03)		(0.02)	(0.02)	
Years of experience	12.40	0.50	996	−0.03	0.01	0.26
	{10.02}	(0.66)		(0.02)	(0.02)	
Monthly profits	6,804	−318	993	0.05	−0.01	0.42
	{9,274}	(567)		(0.03)	(0.06)	
Monthly revenue	34,676	−2,271	993	0.11	0.24	0.76
	{73,690}	(4,513)		(0.13)	(0.41)	
Monthly expenditures	27,932	−2,115	995	−0.12	−0.23	0.79
	{69,240}	(4,287)		(0.12)	(0.37)	
Monthly wage bill	1,384	115	1,004	−0.08	−0.01	0.08
	{3,752}	(251)		(0.04)	(0.03)	
Has employees (=1)	0.36	0.00	1,003	0.04	0.05	0.50
	{0.48}	(0.03)		(0.02)	(0.02)	
Number of employees	0.65	−0.06	1,003	0.03	−0.06	0.02
	{2.18}	(0.12)		(0.02)	(0.03)	
Size of previous ABA loan	7,512	6	1,004	−0.02	0.02	0.13
	{4,636}	(248)		(0.02)	(0.02)	
Have a loan other than ABA (=1)	0.09	−0.03	1,004	0.03	0.00	0.46
	{0.28}	(0.02)		(0.04)	(0.02)	
Value of other non-ABA loans	2,769	−1,294	1,004	−0.03	0.00	0.47
	{10,523}	(581)		(0.04)	(0.02)	
p-value for joint test of standard variables		0.452	991			

Notes: Column 1 reports the control group mean with standard deviations in braces. Column 2 reports the coefficient on a treatment indicator from a regression of each row on that treatment indicator and strata fixed effects. Nonbinary variables are winsorized at the top 1 percent level. Columns 4 and 5 report coefficients from a single fully interacted regression of a binary for take-up of the loan on each variable listed in the rows interacted with an indicator for control (coefficients in column 4) and an indicator for treatment (coefficients in column 5). For columns 4 and 5, nonbinary variables are standardized to simplify interpretation. Column 6 reports the p-value for testing the coefficients from column 4 to its relevant counterpart in column 5. Standard errors in parentheses.

employees, and, for those that have any employees, the average number is 1.75). Monthly profits average about £E6,800, equivalent to US$525. The average size of a firm's previous loan from ABA was about £E7,500. At the time of the baseline survey, only 9 percent were borrowing from other sources. The average outside debt was about £E2,800, but this number is driven by a few large outliers.

Column 2 reports the baseline difference between the treatment group and control group, after controlling for the stratification variables (i.e., loan officer fixed effects). None of the differences is statistically significant, and a joint test of significance, reported at the bottom of the table, yields a p-value of 0.461.

Online Appendix Table 1 reports summary statistics and balance verification for a subset of the psychometric and cognitive variables. The table reports on six key psychometric variables that correlate strongly with treatment effects (we discuss below how we identify these six questions) as well as the cognitive and risk aversion questions. We find no statistically significant differences between treatment and control groups. In a joint test, we find that the entire set of 50 psychometric variables collected at baseline as well as the cognitive and risk questions do not predict treatment status (p-value $= 0.54$).

We were successful in reaching most of our sample for the 20-month survey (96.3 percent response rate) and the 30-month survey (95.1 percent response rate). Column 1 in online Appendix Table 2 shows that there is no evidence of differential attrition when we regress a binary for attrition on the treatment indicator directly. Columns 2–5 regress attrition on a large number of baseline covariates (columns 2 and 4) and those covariates interacted with treatment (columns 3 and 5). A joint test of significance on the coefficients of the interacted characteristics produces a p-value of 0.49, providing confidence that differential attrition is not an issue for our analysis.

II. Average Impacts: Intent to Treat Estimates

We use a standard ANCOVA regression framework to estimate average intent-to-treat effects, pooling our two rounds of follow-up data. Specifically, we estimate

$$Y_{it} \ = \ \beta_1 T_i + \beta_2 Y_{i0} + \delta_{LO} + \lambda_t + \varepsilon_{it},$$

where we have up to two observations per person (one for each of the two follow-up surveys), Y_{it} is the outcome of interest at time t, and β_1 is the coefficient of interest on a binary variable that is equal to one for individuals assigned to the treatment group. We include the baseline value of the dependent variable as a control when available as Y_{i0}, loan officer fixed effects as δ_{LO}, and an indicator variable for the second survey round as λ_t. We include loan officer fixed effects since our randomization stratifies at that level, and we cluster standard errors at the individual level.

A. *Borrowing at ABA* (*First Stage*)

Table 2 reports how borrowing differed by treatment assignment. Of the control group, 76 percent claimed their loans, while 85 percent of the treatment group did so. As intended, we find that the average loan size in the treatment group is more than double the loan size in the control group. Standard loan terms were 12 months, but borrowers were allowed to request longer to repay, and requests were considered by the central credit committee. Treatment group borrowers' loan terms were 50 percent longer on average (19.7 months versus 13.2 months on average). Thus, while monthly payments for the treatment group were larger than for the control group, the gap is around 40 percent rather than 100 percent.

Figure 1 shows how outstanding debt from the lender differs across treatment and control groups over time, extending three years after dispersal of the treatment

TABLE 2—FIRST STAGE: EFFECT OF RANDOMIZED TREATMENT STATUS ON ABA BORROWING

Borrowing outcomes (first stage)	Control mean (1)	Treatment–control difference (OLS) (2)	Observations
Loan take-up (=1)	0.76	0.09	1,004
	{0.43}	(0.02)	
Loan size (£E)	9,735	10,768	1,004
	{9,423}	(884)	
Loan term (sample: take-up = 1), months	13.21	6.49	810
	{3.68}	(0.34)	
Size of monthly loan installments	821	314	1,004
	{940}	(68)	
ABA outstanding debt ~20 months after randomization	4,932	973	1,004
	{8,882}	(615)	
ABA outstanding debt ~30 months after randomization	4,243	1,285	1,004
	{7,433}	(603)	
Took another loan after end of experimental loan (=1)	0.48	−0.06	1,004
	{0.50}	(0.03)	

Notes: This table reports the intent to treatment estimates of the impacts of the larger loans. Column 1 reports the control group mean with standard deviations in braces. Column 2 reports the coefficient on a treatment indicator from a regression that includes strata fixed effects. Standard errors in parentheses.

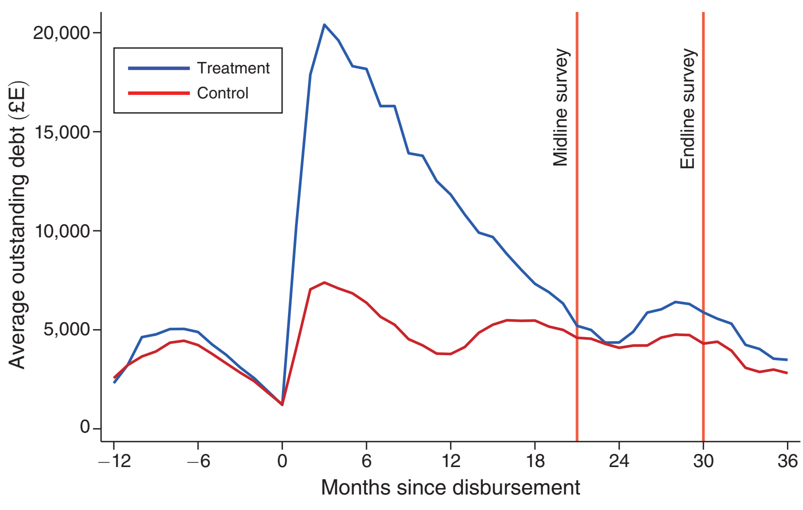

FIGURE 1. OUTSTANDING DEBT AT ABA OVER TIME

loans. It shows that treatment and control had very similar levels of debt in the year preceding the experiment. After the experiment, outstanding debt climbs for the control group but climbs dramatically higher for the treatment group. Given the differences in the length of the loan, the figure shows how the control group's outstanding debt falls around the 12-month mark as their experimental loan is repaid and then increases again, showing another small dip at 24 months. For the treatment group, outstanding debt is at its lowest level at around 24 months after disbursement, which was roughly when many pay back their experimental loan, and then increases afterward.

We consider what predicts take-up in columns 4 and 5 of Table 1. We run a single fully interacted regression where we regress take-up on baseline variables interacted with control and an identical set interacted with treatment. In control, women are more likely to take up the loan, as are those with a registered business, higher monthly profits, and lower spending on labor. In treatment, women continue to take up the loan more often, as do people with some employees. Column 6 reports the *p*-value for equality of the coefficients across treatment and control. We find that wage bill and number of employees differentially predict take-up in treatment and control, and so the composition of those that took the loan will be slightly different between the two groups. Online Appendix Table 1 repeats this analysis but with the cognitive and psychometric measures and finds limited evidence of any differences across treatment and control groups.

B. *Average Impacts on Loan Repayment and MFI Profits*

Using administrative data from the lender, we are able to assess how borrower repayment behavior differed between the treatment and control groups. Table 3 reports that every individual eventually repaid their loan, implying that the concerns over higher default rates were far from realized. Despite full repayment, many borrowers struggled to make all of their payments on time, with treatment firms faring worse. We find that 76 percent of the control group had a perfect repayment record, compared to only 63 percent of the treatment group. Similarly, the average number of days with a payment overdue was 12.7 for control and 26.8 for treatment.[18]

The increase in late payments naturally leads to an increase in payment penalties, which allows us to assess the impact of these larger loans on the lender's bottom line. Table 3 calculates total penalties paid by borrowers in the first 24 months after disbursement. This time frame is long enough to account for the fact that some control firms may have taken two loans in the time a treatment firm had only one. Over those two years, control firms average £E131 in penalties, while those in treatment average £E324—an increase of £E193. This increase in penalties paid is much greater than the lender's opportunity cost associated with that money during the time the borrowers are late in making their payments. As indicated in the final row of Table 3, the treatment group has, on average, £E29 more outstanding on their loan repayment on any given day. This is money that the lender could not lend out. Multiplying this number by a daily interest rate gives the total forgone revenue per day for ABA from loans outstanding. Even if the daily interest rate was 0.5 percent (which is far too high), that would mean only £E0.15 per day, or a total of £E106 over 24 months. In contrast, the total late fees collected by ABA over 24 months was £E193 higher in the treatment group, indicating that ABA earned more in late fees than it lost in forgone lending revenue.

[18] As discussed above, the treatment group had loans that had longer terms of maturity on average, and so we also consider the average number of late days per month of the loan and find that it is 47 percent higher in treatment relative to control, p-value < 0.01.

TABLE 3—EFFECT OF RANDOMIZED TREATMENT STATUS ON REPAYMENTS TO ABA

Repayment behavior	Control mean (1)	Treatment–control difference (OLS) (2)	Observations
Eventual repayment	100%	0	1,004
	—	—	
Perfect repayment of each monthly payment	0.76	−0.13	1,004
	{0.43}	(0.03)	
Total days late over the loan cycle	12.51	14.22	1,004
	{25.67}	(2.44)	
Total penalty within first 24 months	131	193	1,004
	{288}	(46)	
Daily value of late payments over the first 24 months	37	29	1,004
	{59.4}	(7.3)	

Notes: This table reports the intent to treatment estimates of the impacts of the larger loans on repayment behavior. Column 1 reports the control group mean with standard deviations in braces. Column 2 reports the coefficient on a treatment indicator from a regression that includes strata fixed effects. Standard errors in parentheses.

To determine whether or not the larger loans were more profitable from the lender's side, we also need to consider the potential increased costs of loan officer time dedicated to following up with the borrowers to ensure eventual payment. Loans officers and senior management report that loan officer time did not change significantly due to the offsetting forces of decreased administrative load from fewer larger loans and additional time spent focusing on following up with the borrowers of the larger loan. Both loan officers and senior management grew to view the program as a success from a business perspective, and they expanded it throughout the entire organization after the completion of the experiment. Hence, we conclude that the larger loans were profitable for ABA. It is important to note that these results may not generalize to contexts where the potential penalty for default is not as high.

C. *Average Impacts on Primary Outcomes*

We report the average impact of treatment on six primary outcomes in Table 4, panels A and B, column 2. Profits (panel A) increase by £E1,294 a month. This represents an 8 percent increase (SE = 7.5 percent) relative to control, which is positive but not statistically significant. Impacts on revenues and expenses are similar, with an increase of £E5,312 (13.8 percent increase, SE = £E4,446) and £E4,958 (17.6 percent increase, SE = 3,522), respectively, neither of which is statistically significant at conventional levels.[19] We also break out expenditure on employment in particular ("Wage bill") and do not find a statistically significant increase (£E118, SE = 242, a 6.2 percent increase relative to control). We also generate a rough measure

[19] Our profit measure asks business owners to directly report their aggregate profits, instead of asking expenditures and revenue separately and subtracting. This has been shown to be preferable due to the potential for revenues and expenditures to be timed differently in a given month (De Mel, McKenzie, and Woodruff 2009).

TABLE 4—PRIMARY IMPACTS ON ENTERPRISE OUTCOMES: ITT AND GATES ESTIMATES

	Control group mean	Treatment–control difference (OLS)	Group average treatment effects (GATES)			
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Profits						
Profits	15,649	1,294	−8,180	1,840	5,325	8,611
		(1,180)	(3,584)	(1,872)	(3,078)	(3,742)
	Control group mean	Treatment–control difference (OLS)	Conditional group average treatment effects (CGATES)			
Panel B. Other primary outcomes						
Revenues	38,339	5,312	−43,058	2,749	11,971	50,942
	0	(4,446)	(9,163)	(9,119)	(6,113)	(9,232)
Expenses	28,190	4,958	−30,943	7,107	5,771	38,770
	0	(3,522)	(7,567)	(6,854)	(5,213)	(7,897)
Wage bill	1,913	118	−1,596	−988	633	2,678
	0	(242)	(540)	(523)	(475)	(547)
Total factor productivity	−0.02	0.04	−0.49	−0.08	0.12	0.62
	0	(0.05)	(0.10)	(0.09)	(0.10)	(0.10)
Household expenditure	4,770	446	−440	−162	−1	2,182
		(211)	(450)	(415)	(383)	(453)
	Mean	Difference (OLS)	Conditional group average treatment effects (CGATES)			
Panel C. Secondary outcomes						
Has a business	0.96	0.00	−0.07	−0.02	0.04	0.05
		(0.01)	(0.02)	(0.02)	(0.02)	(0.02)
Number of employees	2.06	−0.14	−1.72	−0.92	−0.06	1.58
		(0.26)	(0.55)	(0.41)	(0.83)	(0.39)
Value of assets	192,733	−1,932	−105,322	12,675	−63,443	131,958
		(28,356)	(48,942)	(55,721)	(64,271)	(62,982)
Mental health index	0.01	0.00	−0.20	−0.03	0.13	0.09
		(0.05)	(0.10)	(0.10)	(0.10)	(0.10)
Physical health (out of 5)	2.18	0.04	0.11	0.00	0.07	−0.05
		(0.05)	(0.09)	(0.10)	(0.10)	(0.10)
Total penalty within 24 months	131	193	235	261	176	101
		(44)	(129)	(77)	(100)	(79)
ABA loan at 30 months	0.43	−0.04	−0.10	−0.02	−0.06	0.01
		(0.03)	(0.07)	(0.07)	(0.06)	(0.07)

Notes: Column 2 reports the ITT estimate from a regression of the outcome variable (row) at follow-up on a treatment indicator. The regression includes controls for strata, survey round, and the baseline value of the outcome when available. Standard errors clustered at the indvidual level. Columns 3–6 report results from a two-step process. First, using the random forest algorithm with psychometric, risk, and cognitive data, and treatment status, we predict individual treatment effects on business profits. Second, we report the coefficients from a regression of the outcome variable (row) on treatment status interacted with indicator variables associated with which quartile of predicted individual treatment effects (columns) the borrower was assigned to in the first step. For panel A, this second step is done for profits, the same outcome used to estimate the individual treatment effects. For panels B and C, this second step is done for other outcomes, but still the quartile assignment is based on predicted treatment effects on profits. All regressions include controls for predicted profits if in treatment, predicted profits if in control, strata fixed effects, survey round, and cluster standard errors at the individual level.

of total factor productivity (TFP) utilizing endline revenues and find an increase of 0.04 standard deviations, which is also not statistically significant.[20]

Our final primary outcome is household expenditure. There, we find an increase of £E446 (SE $=$ 211), a 9.4 percent increase relative to the control group, with a standard p-value < 0.10.[21] The statistical significance of this result does not survive correcting for multiple hypothesis testing using Anderson (2008). All together, these results provide suggestive evidence that the treatment could be leading to increases in business performance and household outcomes on average.

Table 4, panel C, column 2 reports the average impacts of the treatment on seven additional outcomes. Three are important business outcomes: whether the borrower still has a business, the total number of employees, and the value of assets. We find no statistically significant average impacts across any of these variables. We also consider two downstream measures of health, an index of mental health questions and a self-report on physical health, and again find null average impacts. We also include two measures related to loan performance. We reproduce the impacts on total late fees in the 24 months after the experimental loan, which increases on average. We also consider whether or not individuals have an outstanding ABA loan 30 months after the experimental loan and find a decrease of 4 percentage points in the treatment group, but this is not statistically significant.

III. Heterogeneous Treatment Effects

While the average treatment effects suggest a small positive effect of larger loans on business outcomes, it is possible that these average impacts hide important heterogeneity and hence that better allocation of credit could lead to larger average treatment effects. We collected the psychometric data to explore this possibility, given previous work that suggests that these types of data could be predictive of performance (Klinger, Khwaja, and LaMonte 2013; Arráiz, Bruhn, and Stucchi 2017). To assess heterogeneity, we follow Chernozhukov et al. (2023), which provides a strategy for determining whether there is significant treatment effect heterogeneity in randomized experiments where the set of covariates is high dimensional and there is no clear ex ante hypothesis on which to base a pre-analysis plan.

Because the main aim of the experiment was to increase the profitability of firms, we first look at heterogeneity in the average treatment effect (ATE) for profits. In doing so, we follow Chernozhukov et al. (2023) closely, and we give a brief description of the method for completeness. We follow a slightly different approach for other outcomes, which we discuss below.

[20] Since we do not have assets at baseline, we estimate TFP based on a Cobb-Douglas production function that only includes labor and assets in a specification where we take the residual of a regression of log revenue on log assets and log wage bill using endline data and then standardize it. We recognize that there are endogeneity concerns with this procedure and that our approach does not meet the standards of the current literature on estimating the production function (see, e.g., De Loecker and Syverson 2021 for a review of current methods). Unfortunately, we lack the data to implement any of the control function of dynamic panel methods that are the current standard. Given this, our results must be treated with caution. We use a revenue-based measure of productivity following Atkin, Khandelwal, and Osman (2019), which shows that revenue-based measures of productivity (TFPR) perform better than quantity-based measures (TFPQ).

[21] The impact on household expenditure is statistically significant using standard inference. This is consistent with the average impact on profits, with the difference in significance coming from the difference in noise in these two variables, which likely reflects some consumption smoothing.

To estimate heterogeneity in the treatment effect for profits, we first split the sample into a "training set" and a "testing set." Throughout, we use 50/50 splits. Using the training set only, we train a machine learning (ML) method to generate a "control" effect $B(Z_i)$ (i.e., the expected outcome for firms with covariates Z if they were assigned to control) and predicted treatment effect $S(Z_i)$, where Z_i denotes the full set of covariates used to predict heterogeneity for subject i. Below, we show results where we alter the set of covariates used. Any ML method could be used, but we use four default options (elastic net, neural net, random forest, and gradient boosting) and then take the one with the highest prediction score.[22] Note that because we utilize all four ML methods and choose the one with the highest prediction score, we utilize a conservative Bonferroni correction in our estimates and multiply all of the p-values by four, in line with Chernozhukov et al. (2023). In all cases, we use the implementation of these methods from the R package "caret."

With the estimates $B(Z_i)$ and $S(Z_i)$ in hand, we then undertake two analyses using *only* data from the testing set. First, we estimate the regression

$$Y_i = \alpha \times X_i + \beta_1 \times T_i + \beta_2 \times T_i \times S(Z_i) + \epsilon_i, \tag{1}$$

where X_i is a set of covariates that includes $B(Z_i)$ and T_i is an indicator for treatment group.[23] Our primary use for this specification is to test the null hypothesis of no heterogeneity, $\beta_2 = 0$.[24] Second, we split the testing sample into quartiles of predicted treatment effect using $S(Z_i)$ and estimate the regression

$$Y_i = \alpha \times X_i + \sum_{j=1}^{4} \gamma_j \times T_i \times 1(S_i \in I_j) + \eta_i, \tag{2}$$

where I_j is the set of firms in the jth quartile.[25] γ_j measures the "sorted group average treatment effect" (GATES) for each quartile and is the key measure that we use to understand how treatment effects differ across well-defined groups.

The key contribution of Chernozhukov et al. (2023) is to show how to get theoretically correct inference for these analyses, and, again, we follow their approach. We repeat the split into training and testing sets 100 times (each with a different randomly chosen split) and run the analyses in (1) and (2) for each split. This process produces estimates of the key parameters β_2 and γ_j for each of the 100 splits, as well as the associated confidence intervals, standard errors, and p-values. For the parameter estimates, we report the *median* from the 100 runs. For a $1 - \alpha$ confidence interval, we report the median of each boundary of a $1 - \alpha \times 2$ confidence interval from each split. For hypothesis tests in equation (1), we state that a hypothesis is significant at the α level if the median p-value is less than $\alpha/2$. The use of $\alpha/2$ in the hypothesis tests and confidence intervals corrects for sample splitting. As mentioned

[22] This is defined as $|\hat{\beta}_2|^2 \widehat{Var}(S(Z))$, where β_2 is defined in equation (1).

[23] The treatment assignment is included as the treatment binary minus a propensity score associated with treatment assignment. The propensity score is constant due to the randomized treatment assignment. The individual treatment effect $S(Z_i)$ is included as a deviation from its mean.

[24] $\beta_2 = 0$ if there is no heterogeneity or the ML prediction $S(Z_i)$ does not capture that heterogeneity. Hence, this test is of a joint hypothesis, that there is heterogeneity and that the ML methods can detect it using the covariates that we have.

[25] Again, the treatment assignment is included as the treatment binary minus a propensity score associated with treatment assignment.

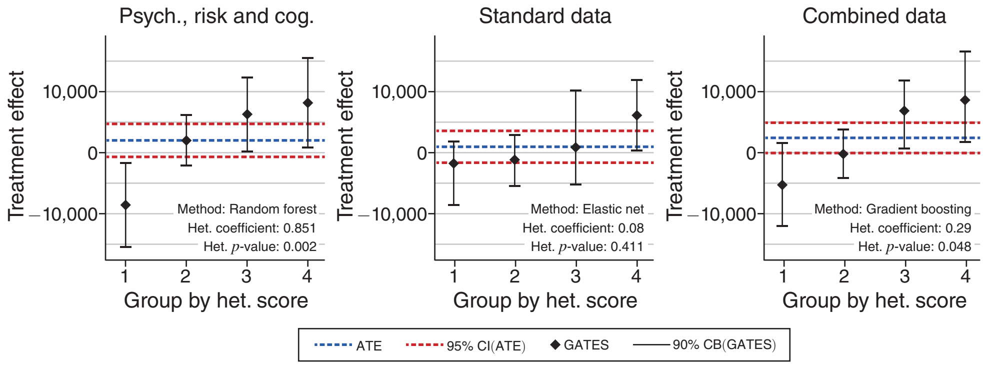

Figure 2. Heterogeneous Treatment Effects for Profits

above, we implement a Bonferroni correction by multiplying p-values by four due to the initial test of four machine learning prediction methods.[26]

Results are presented in Figure 2 and Table 4, panel A, columns 3–6. Figure 2 shows GATES for the quartiles using three different definitions of Z_i: just the psychometric/cognitive data, just the standard data, and all data (both psychometric/cognitive data and standard data). The first point to note is that there is strong evidence of heterogeneity. Using just the psychometric/cognitive data,[27] we reject the hypothesis of no heterogeneity in treatment effects ($\beta_2 = 0$) and estimate $\beta_2 = 0.85$ with a p-value of 0.002. This coefficient being so close to one shows that the model does a very good job at predicting treatment effects accurately.[28] Concentrating on the estimates using the psychometric/cognitive data, we find in Table 4, panel A, columns 3–6 that those in the most positively affected group see about an £E8,600 increase in monthly profits and that this impact is significantly different from 0 at the 5 percent level. At the other extreme, those in the least affected quartile *lose* around £E8,200 per month in profits because of the treatment, an effect that is also statistically significant at the 5 percent level.[29] To put these coefficients in perspective, mean profits in the control group in our sample are about £E15,650 per month, which means that the positive gain for the top quantile is about 55 percent of the control mean, while the loss for the least affected group is about 52 percent of the control mean.[30] As a robustness check, we replicate the analysis but replace the

[26] We note that this procedure is quite conservative, and future researchers may consider departing from the $\alpha * 2$ adjustment for sample splitting when using a large number of splits.

[27] When we refer to "psychometric/cognitive data," we include all 50 psychometric measures, as well as our measures of risk aversion and cognitive ability. When we implement the analysis without the risk aversion or cognitive ability measures, our estimates of heterogeneity are just as strong.

[28] We also find statistically significant evidence of heterogeneity using other machine learning algorithms included in the Chernozhukov et al. (2023) method, but we focus on the estimates from the method with the highest prediction performance.

[29] It may appear that these impacts are quite symmetric, in the sense that top performers do about as well as poor performers do badly. This is in part an artifact of our emphasis on the top and bottom groups (which we have more power to distinguish). Looking more closely at the distribution, it seems plausible to argue that there are a small group of firms that do badly with the loan but that most other firms do reasonably well, earning a positive return, although not statistically significant for the middle two GATES groups.

[30] This increase in profits suggests a monthly rate of return of nearly 11 percent on the additional capital for top performers, about twice the average rate of return found in De Mel, McKenzie, and Woodruff (2009). Given that these returns are for those who are most capable of utilizing the larger loan, this seems reasonable.

profit variable with a random variable that has the same mean and standard deviation as the actual profit variable. In that case, we fail to reject the null hypothesis, with an estimated β_2 of 0.11 and a p-value of 0.80, and no statistically significant difference between the top and bottom groups.[31]

It is important to note that the psychometric/cognitive data do a much better job of capturing heterogeneity in treatment effects than do the standard data, and the inclusion of the standard data adds noise.[32] When we use only the standard data, the estimated p-value increases to 0.411 (so that we cannot reject the joint hypothesis of zero heterogeneity or that the standard data cannot detect the heterogeneity). When we utilize all of our data, the p-value associated with the heterogeneity is 0.048, and the difference in GATES between the top and bottom groups is about £E12,000. In contrast, when we restrict our data only to the psychometric/cognitive questions, we rule out homogeneity with a p-value of 0.002 and a bottom versus top group difference of over £E16,000. This implies that it is primarily the psychometric data that are allowing us to predict the heterogeneous treatment effects. This result also implies that these machine learning techniques cannot fully differentiate signal from noise, as evidenced by the decrease in performance when using all of the data relative to only the psychometric/cognitive data. While the algorithms do help deal with the high dimensionality of the data, there is still an important role for researchers to identify which variables to include, and in what form (Mullainathan and Spiess 2017).

These estimates show that there is a subset of the sample who are using the loan to greatly increase profits and another group where profits decrease in response to the larger loan. Critically, these groups are not generated in a way that overfits the data to maximize heterogeneity; rather, they are produced through a method that utilizes baseline data and split-sample validation. This means that the individual treatment effects (ITEs) estimated for each person do not use any of the data on that individual's performance. Hence, it is not equivalent to grouping individuals who may have realized a negative/positive shock in their business and splitting up the sample based on realized outcomes. Instead, this categorizes people based on their baseline characteristics and the algorithms' predictions about how they would perform with the larger loan based on the performance of other people who are observationally similar at baseline.

The results indicate that there is a "type" of borrower who will be able to improve their business with larger loans and a "type" of borrower who will actually hurt their business when provided with a larger loan, relative to a smaller loan. This is particularly striking given the several layers of review that were already built into the selection of the sample. In addition to the borrowers requesting the loan on their own behalf, the loan officers needed to approve the individual getting a larger loan, the borrower had to have successfully completed three previous loans, and a central credit committee needed to review their case before they were randomized. All of the links across that chain were intended to weed out individuals who would not

[31] We also replicate the analysis while dropping individuals who did not take out a loan in either treatment or control and find very similar results, suggesting that the observed heterogeneity is not driven by differential loan take-up rates.

[32] We outline all of the variables that are used in our "standard data" in the prediction exercise in the online Appendix. This includes, among other variables, all of the firm's baseline characteristics as well as prior loan size and performance and the sector that the firm is in.

benefit from the loan, but even in this highly selected sample, there is significant heterogeneity and a group of borrowers who are negatively affected. We explore this further in the following section.

These results are in line with earlier work on microcredit that shows evidence of important heterogeneity in impacts. For example, Banerjee et al. (2019) finds that microcredit leads "gung ho entrepreneurs" to more than *double* profits relative to those in control. Crépon, Komi, and Osman (2024) also provides evidence that credit can double profits and is particularly heterogeneous as well. Related, in a seven-site meta-study employing Bayesian hierarchical modeling, Meager (2020) finds large positive impacts for a subset of borrowers *and* evidence of negative returns to credit for some borrowers in two of the seven sites (Mongolia and Ethiopia).

Conditional Group Average Treatment Effects.—Having found heterogeneity with respect to profits, in this section, we explore whether firms that saw an increase in profits also saw changes in other related outcomes. Our aim is to shed some light on mechanisms and to validate the plausibility of our results. For example, understanding whether the top performers, those with the highest ITEs on profit, also increased revenue, wages, etc., can build insight regarding the likely path toward increased profits. For example, do profits increase via an expansion of the business or via a shrinkage of unprofitable inputs? The former implies a likely increase in wage bill; the latter implies a reduction. While the method from Chernozhukov et al. (2023) allows for assessing heterogeneity on any variable, it does not directly allow for determining whether heterogeneity in profit treatment effects translates into heterogeneity in other variables. Hence, we slightly extend the GATES procedure from Chernozhukov et al. (2023) to estimate conditional sorted group average treatment effects, or "CGATES."[33]

The CGATES procedure follows the normal GATES procedure closely. For each of the sample splits (into training and testing), we train our ML method on the training data and calculate $S_i = S(Z_i)$, the predicted individual treatment effect for i, for each individual in the *testing* group. At the end of this procedure, we take the median of these individual treatment effects for each i to get a person's predicted treatment effect *for profits* and allocate individuals into quartiles.[34] We then run the same specification as in equation (2) but change the Y variable to be the other outcomes variables of interest. In particular, we estimate

$$Y_i = \alpha \times X_i + \sum_{j=1}^{4} \gamma_j \times T_i \times 1(S_i \in I_j) + \eta_i, \tag{3}$$

where I_j is an indicator for individual i being in the Nth quartile ITE group for *profits*, which we refer to as the Nth CGATES group.[35]

[33] This analysis is similar to what is implemented in Bertrand et al. (2021).

[34] The groups that are produced from this method will differ on baseline characteristics (as we will show below), but the key is that the predicted treatment effect S_i is balanced between treatment and control within each subgroup.

[35] Again, the treatment assignment is included as the treatment binary minus a propensity score associated with treatment assignment, and the X vector of controls includes the predicted values of $B(\cdot)$ and $S(\cdot)$ for profits. Since group allocation is different in each split of the estimation, this implies that the threshold ITE for allocation to different groups could be different across splits. For this reason, we generate the group for CGATES by ranking people based on the average group an individual was allocated to across all splits. When we instead rank based on the median ITE across splits, our results are qualitatively unchanged, but we prefer doing it based on average groups because this is immune to concerns about different average ITEs across splits.

We are interested in γ_j, which is the average treatment effect for individuals who were allocated to the four different CGATES groups. The main difference between this method and the Chernozhukov et al. (2023) method is that the group assignment is based on profits and not on the other outcomes of interest that we analyze. This allows us to assess how impacts differ for individuals who increased their profits in response to the larger loan and those who decreased their profits. We also take the predictions across the 100 simulations and then allocate individuals to groups, instead of taking the median coefficient from regressions across the 100 splits of the data. We then apply a Bonferroni correction by multiplying p-values for each coefficient variable by four as above (since we are using estimates from the strongest of four machine learning algorithms) and then further correct for multiple hypothesis testing in the table using sharpened q-values as in Anderson (2008).

Table 4, panels B and C, columns 3–6 report the results of these regressions. Panel B shows impacts on our primary outcomes other than profits. Column 3 reports the treatment effect for individuals who were in the first CGATES group (those in the lowest quartile of individual treatment effects for business profits). Columns 4–6 follow the same format, reporting the treatment effect estimates for individuals who were in the second, third, and fourth CGATES groups. The individuals in each group are held constant across all of the regressions in Table 4, panels B and C since we are conditioning on the ITE group for *profits* and not for the ITE groups optimized for the outcomes presented. Many of these outcomes are highly correlated, and so we expect results to be similar to the impact on profits, but this exercise allows us to better understand the mechanisms through which profits are increasing.

The first row in panel B shows that those in the highest ITE quartile for profits also have a large increase in revenues, with an increase of £E50,942 (SE = 9,232), while those in the lowest CGATES group see a revenue *decrease* of £E43,058 (SE = 9,163). A similar pattern holds for the other business outcomes in the subsequent three rows—expenditures, spending on wages, and estimate TFP: there are large statistically significant gains for those in the top CGATES group and losses for those in the bottom CGATES group. For example, those in the top CGATES group increase their wage bill by £E2,678 per month (SE = 547), while those in the bottom CGATES group decrease their wage bill by £E1,596 (SE = 540).[36] We also find large differences in estimated TFP in the top and bottom quartiles. The final row in panel B shows that the heterogeneous effects in business outcomes are also reflected in reported household expenditure. The top group experiences a statistically significant £E2,182 increase in expenditure, while the bottom group sees a decrease of £E440, although this decrease is not statistically significant.

Panel C continues our analysis of heterogeneous treatment effects showing estimated treatment effects for each CGATES group on additional outcomes. The table shows that for those in the bottom CGATES group, the larger loan decreased the likelihood that they were able to keep their business open by 7 percentage points. The loan also led to an increase in the number of employees by 1.58 employees for the top group and a decrease of 1.72 employees in the bottom group. The point

[36]The decrease in business expenses is larger than the average value in the control group. This is likely due to nonlinearities in the size of the control group in the follow-up surveys by treatment quartile. It's worth noting that there are no differences in *baseline* values of expenses.

estimates for changes in the value of assets are large but not precisely estimated.[37] Estimates of the heterogeneous impacts on mental health suggest a similar pattern, with increases for those in the top profits group and decreases for those in the bottom groups. It is worth noting that there is a decrease in reported mental health of 0.20 standard deviations for those in the bottom quartile. This decrease is large, and, while not statistically significant, it suggests there could be an important mental toll that comes from a failing business.

Together, these results show that top performers are succeeding through business expansion and not through cutting costs and slimming down. Revenues, expenditures, and their wage bill all increase. There is suggestive evidence that the value of assets also grows. This contrasts sharply with poor performers, who do not simply run up expenses while keeping revenues constant but actually see revenues and asset values drop, suggesting that as a group, they "swung and missed" by pursuing strategies that had the potential to increase revenues but also the potential to decrease revenues.

While we emphasize the idea that our results suggest misallocation of credit, an alternative interpretation is that expanding access to credit causes bad businesses to close and so acts to reallocate capital, even if ABA cannot find the top performers. Unfortunately, we are not able to provide direct evidence on this hypothesis, given the small number of firm closures and limited time frame. The negative effects on entrepreneur mental health, however, suggest that this approach would have a negative human toll.

We also explore heterogeneity in borrowing behavior. We find some evidence that those in the bottom groups pay more in late fees, but this result is not statistically significant. We also find that those in the lowest group are less likely to be borrowing after 30 months, likely because many have closed down their businesses.

IV. The Quality of Allocations

The heterogeneity of impacts discussed above suggests substantial scope for the allocation of large loans to affect the impact of credit availability and the aggregate profitability and productivity of firms borrowing from ABA. We now discuss whether, in the absence of psychometric data and machine learning techniques, ABA would be in a position to make efficient allocation decisions. As noted above, studying the allocation problem requires providing large loans to firms that would not normally receive large loans in the "business as usual" case in the absence of our intervention. We hence use two different proxies that we believe provide a good sense of which firms would likely have been allocated larger loans if an expansion like this one had happened normally.

[37] While we expect increasing asset values to contribute to the profit increase, it is not necessary for assets to increase for a business to become more profitable. It could be that they used some of the funds on advertising, training, or other business support services that don't directly show up in asset values.

A. *Firms That Perform Well with Small Loans*

A natural theory of loan allocation is that the lender will perceive firms that perform well with smaller loans as high-quality entrepreneurs and hence will be more likely to extend larger loans to them. This would also be an optimal loan allocation strategy in models such as Banerjee and Moll (2010), which posits a constant firm-specific productivity term that multiplies a decreasing returns to scale production function. According to this theory, the allocation of large loans is likely to be relatively efficient if top performers are among those firms that see the greatest firm growth when given a small loan, and inefficient if top performers perform relatively badly with small loans. Evaluating the quality of allocation in this way has the advantage that it moves beyond the specific allocation decisions of ABA to evaluate the effectiveness of this simple allocation rule on the firms within our sample, which we believe to be relatively representative of SMEs in Egypt.

Table 5 presents evidence consistent with a propensity toward misallocation of larger loans if our basic theory is correct. Each column in the table is a regression in which we regress a measure of firm performance on a standardized measure of our predicted individual treatment effect. The top performers have high values for this index. Panel A has baseline to endline change in profits as the outcome measure of firm performance, while panel B has endline TFP.[38] Columns 1 and 2 show that our top performers do badly when they are placed in the control group: they are the firms that see the smallest baseline to endline increase in profits and also the firms that have the smallest endline TFP. In contrast, columns 3 and 4 show when the top performers are placed in the treatment group, they see the largest increases in profits and have the highest endline TFP. The availability of credit leads to a strong reversal of fortune. Overall, these results suggest the entrepreneurs who benefit most from the loan make gains because they would have done poorly in the absence of the loan. Those who lose from the loan do so because they would have done well without it. It also implies that misallocation will ensue if a lender pursues a strategy of targeting large loans at those firms that perform well with small loans.

B. *Loan Officer Perceptions*

Our second strategy for understanding likely misallocation is more strongly tied to ABA's institutional structure. As noted above, loan officer incentives strongly favor minimizing default, and loan officers have a great deal of control over which firms get loans. This implies that if loan officers believe top performers will see relatively low defaults, then large loans will be well allocated. Simultaneously, it also implies that loan officers will misallocate large loans if they perceive top performers to be likely to default. In favor of the reasonableness of this assumption, we asked loan officers to list their most important considerations when deciding whether or not to provide a loan to a client. Repayment ability was listed as the top reason by 70 percent, with another 20 percent listing it as the second-most important reason.

[38] We do not have the data necessary to compute baseline TFP.

Table 5—Correlates of Enterprise Profits and Total Factor Productivity

Sample frame:	Control only		Treatment only		Full sample	
Outcome (£E/month):	Endline–baseline profits		Endline–baseline profits		Baseline profits	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Business profits						
Standardized predicted treatment effect	−16,396	−16,454	20,201	20,209	−569	−645
	(2,496)	(2,484)	(1,975)	(2,032)	(559)	(566)
Years of education		341		−186		219
		(174)		(260)		(113)
Years of experience		−87		244		28
		(66)		(156)		(70)
Number of observations	949	949	950	950	992	992

Sample frame:	Control only		Treatment only		Full sample	
Outcome = TFP:	(7)	(8)	(9)	(10)	(11)	
Panel B. TFP (follow-up surveys only; asset information not collected at baseline)						
Standardized predicted treatment effect	−0.280	−0.282	0.234	0.229	Treatment	0.045
	(0.059)	(0.058)	(0.039)	(0.038)		(0.052)
Years of education		0.004		0.012	Index	−0.282
		(0.008)		(0.008)		(0.059)
Years of experience		0.009		0.002	Treatment * ITE	0.512
		(0.004)		(0.004)		(0.070)
Number of observations	951	951	953	953		1,904

Notes: Panel A reports results from regressions of profit outcomes on a standardized measure of the predicted individual treatment effect. Regressions include a control for survey round and cluster standard errors at the borrower level. Panel B reports results from a similar regression but with total factor productivity in the follow-up surveys as an outcome.

Table 6, panel A presents results on loan officer perceptions using data collected before the randomization.[39] We asked loan officers to grade borrowers on their "ability to repay the loan" on a 1–10 scale. We asked them the question twice, once about their perception of repayment ability in the case that the borrower received the larger loan and again in the case that the borrower received the smaller loan. We score this as a binary variable that combines the two responses, equal to one if the loan officer perceives the larger loan will be more likely to lead to default than the smaller loan.[40] We compare their responses to these questions for top performers relative to poor performers.

We find that loan officers believe that those in the top quartile of impacts are *more* likely to struggle to repay if provided the larger loan relative to those in the bottom quartile. Loan officers predict that a large loan will increase default for 28 percent of poor performers, and this increases by a statistically significant 18 percentage points for top performers. We see these results as corroborating our analysis above. Loan officers can observe performance with small loans, and they perceive that poor

[39] As noted above, due to logistical constraints, we were only able to start the loan officer survey toward the end of implementation and only have loan officer predictions for the final 29 percent of the loans in our sample. Baseline characteristics for loans we have perceptions about are statistically equivalent to those we don't have perceptions for.

[40] When we implement this analysis but using only the loan officer's assessment of performance on the larger loan, we find similar results.

Table 6—Differences in Baseline Characteristics between Top and Bottom GATES Groups

	Bottom GATES group Mean {SD} (1)	Difference in top group Coeff (SE) (2)
Panel A. Loan officer perceptions		
Perceives large loan will increase default	0.28	0.18
	{0.45}	(0.08)
Perceives large loan will increase firm revenue	0.24	0.14
	{0.43}	(0.07)
Panel B. Standard data		
Demographics		
Age	42.2	−2.99
	{10.3}	(0.91)
Female	0.24	−0.10
	{0.43}	(0.04)
Years of education	7.92	1.38
	{4.78}	(0.41)
Business characteristics		
Registered business	0.36	−0.01
	{0.48}	(0.04)
Monthly revenues	51,902	−2303
	{224,345}	(17745)
Monthly expenses	44,569	−2895
	{221,702}	(17438)
Monthly profits	7,792	−425
	{12,363}	(1029)
Has employees (=1)	0.40	−0.02
	{0.49}	(0.04)
Number of employees	0.73	−0.09
	{2.42}	(0.18)
Size of previous ABA loan	7,660	66
	{4,967}	(436)
Value of other loans	3,758	−2260
	{12,670}	(929)

	Bottom GATES group Mean {SD} (3)	Difference in top group Coeff (SE) (4)
Panel C. Psychometric data		
Cognitive and risk aversion		
Digit span recall	2.20	1.09
	{1.91}	(0.16)
Ravens matrices	1.62	0.29
	{1.44}	(0.13)
Financial literacy score	5.18	1.92
	{3.32}	(0.30)
"Hypothetical % allocated to risky investment"	0.78	−0.13
	{0.65}	(0.05)
"Willingness to take risks (10= fully willing)"	6.79	0.01
	{2.93}	(0.24)
Most predictive psychometric measures		
I tend to act first and worry about the consequences later.	4.38	−0.99
	{0.99}	(0.09)
I can think of several solutions to any problem.	4.68	−0.78
	{0.61}	(0.06)
When I make decisions I usually go with my first, gut feeling.	4.08	−0.25
	{1.27}	(0.10)
I always get things done ahead of time.	4.70	−0.71
	{0.53}	(0.05)
I would work seven days a week if I could.	4.56	−0.41
	{0.89}	(0.07)
I spend a lot of time planning for my future.	3.97	−0.20
	{1.24}	(0.10)
I prefer to have a flexible schedule—I don't like being tied down	4.46	−0.98
	{0.79}	(0.08)
Optimism-related psychometric measures		
In life, failure is not an option.	4.76	−0.78
	{0.51}	(0.06)
When I make a business decision it is almost always the right decision.	4.42	−0.63
	{0.98}	(0.08)
I have always believed I am going to be successful.	4.42	−0.72
	{0.88}	(0.08)
I prefer to focus on opportunities rather than risks.	4.47	−0.52
	{0.91}	(0.07)

Notes: Columns 1 and 3 report the baseline averages of characteristics for the individuals who are in the bottom group of estimated impacts on profits based on the Chernozhukov et al. (2023) method utilizing psychometric data, while columns 2 and 4 report the coefficient on an indicator for being in the top group from a regression of the variable in each row. Regressions only include people in the top or bottom groups. Panel A includes data from a loan officer survey conducted toward the end of implementation and covers 293 borrowers. Business profits, revenues, expenditures, and value of other loans are winsorized at the top 1 percent level. Hypothetical risk investment question: Respondents asked to allocate £E10,000 to either a risky (50 percent chance of doubling/halving) investment or risk-free zero-return investment. Psychometric questions are on a 1–5 scale, with 5 = strongly agree and 1 = strongly disagree. We report the seven psychometric variables that were chosen by both the random forest model and a lasso regression of the estimated individual treatment effect on all psychometric, risk, and cognitive variables, as well as four additional psychometric questions collected at baseline that we manually identify as being related to optimism. All other psychometric questions are shown in online Appendix Table 6. Standard errors in parentheses.

performers do well and believe they will be less likely to default. This implies that, in the absence of some alternative method to identify top performers, large loans would be misallocated. A caveat to this interpretation is that loan officer beliefs were not incentivized, and so they may not have taken the task particularly seriously. We suspect that if this were the case, it would induce measurement error and attenuate coefficients toward zero rather than producing the statistically significant negative effect that we find.[41] Table 4, panel C also suggests there may be a disconnect in observed repayment behavior, with those in the bottom quartile paying more in late fees than those in the top quartile, although these estimates are not statistically different.[42]

While ABA makes its lending choices based on perceived default, an alternative way to allocate credit would be for loan officers to encourage loans to those firms that they believe will benefit most from them. To test whether this strategy would improve allocation, we asked loan officers to predict which firms would see a revenue increase. Table 6, panel A presents results on how these perceptions correlated with treatment effects. We find a marginally significant difference between the top and poor performers, with the sign in favor of efficient allocation: the loan officers believe that 24 percent of poor performers and 38 percent of top performers will experience more revenue growth with the larger than with the smaller loan. While relatively weak, this evidence suggests that allocation based on perceived revenue gains would be more efficient than current practice but not as effective as allocation based on psychometric data. A separate experiment would be required to understand whether a change in the loan officer incentive scheme could remove the misallocation, but the evidence is suggestive that it would.[43] On a more speculative note, the fact that loan officers believe top performers to be more likely to increase revenues implies that they may have the information required to improve allocation. It is an open question whether ABA could redesign incentives to elicit this information.

V. Who Benefits Most from Big Loans?

Our main results are, we think, surprising, suggesting both large heterogeneity in returns to big loans and a reversal of fortune, whereby those who do relatively well with a smaller loan do relatively poorly with a larger loan.

To better understand these results, this section describes how the business and entrepreneur characteristics differ between top and poor performers. Table 6 compares the baseline characteristics of top and poor performers. While the machine learning algorithms are nonparametric in nature, comparing average baseline values across groups can help us better understand what differentiates the two groups.

[41] Online Appendix Table 3, panel A includes a related analysis. We split the sample by loan officer prediction of the impact of the loan on repayment ability and estimate the impact on profits for each group. We find that loan officer perceptions about repayment ability are negatively correlated with impacts on profits.

[42] It is difficult to precisely pin down the source of these loan officer misperceptions. We regress a measure of "accuracy" based on the scores loan officers gave clients regarding their ability to repay the larger loan and the normal loan on a suite of baseline characteristics. We find that the amount of credit a client has from outside sources is negatively correlated with accuracy, which suggests that part of the misperception could be due to how loan officers feel about the risk of overindebtedness.

[43] Online Appendix Table 3, panel B again provides related analysis. It splits the sample based on loan officer perceptions and shows that loan officer perceptions are not very predictive of loan impacts on profits.

In panel B, we consider standard data collected at baseline, split into demographic characteristics and business characteristics. Those in the top group are less likely to be female and are more educated on average by one year. We find no statistically significant differences in baseline business characteristics, including whether the business is registered, baseline profits, revenues, expenditures, or the number of employees.[44] There is also no difference in the size of their previous ABA loan, but those in the top group had about half the amount of loans outstanding with organizations other than ABA. To the extent that non-ABA borrowing is from higher interest rate sources, this result suggests that poor performers are more optimistic about their opportunities and hence more credit hungry.[45]

In panel C, we consider our baseline psychometric measures, split into cognitive and risk aversion and then the more explicit psychometric measures. We find that those in the top group perform better on the digit span recall task and on a battery of financial literacy questions. They also perform better on the Raven's Matrices test, and those in the top group exhibit higher levels of risk aversion, as evidenced by a lower proportion of investment in a hypothetical risky asset relative to the control group.[46]

It is less obvious how to compare the top and bottom groups in their responses to our psychometric questions since we asked 50 questions at baseline and the way these measures are mapped onto specific psychological constructs often relies on researcher discretion. We deal with this challenge in two ways. First, we run a lasso regression where we predict the estimated individual treatment effect while including on the right-hand side all of the psychometric, risk, and cognitive variables.[47] Second, we build a random forest model of the predicted individual treatment effects using the same baseline variables and recover measures of "feature importance" (Athey and Imbens 2019). In essence, this measure tells us how often each variable is used when creating the set of decision trees in the model. We take the seven most predictive measures from each approach and find that both strategies identify the same seven variables as being most predictive, which we then include in Table 6, panel C.[48] Full results from these two approaches are presented in online Appendix

[44] We also considered how the top and bottom groups differ on baseline business sector. We find that those in the top group are more likely to be in retail and services, which are generally seen as less capital intensive than manufacturing and agriculture.

[45] In online Appendix Table 5, we regress the estimated individual treatment effect on baseline covariates and find that previous loan size is positively correlated with it. We include this variable in our "standard data," and it is not sufficient to pick up heterogeneity there.

[46] We ask individuals to imagine that they had £E10,000 that they could invest in an opportunity that would either double their money or halve it, each with a 50 percent probability. They then report the amount of those funds they would invest in the risky asset.

[47] Selection of variables in lasso regressions can be sensitive to the inclusion of highly correlated potential explanatory variables. To overcome this limitation, we utilize the "Puffer Transformation" as developed in Jia and Rohe (2015) and utilized in Banerjee et al. (2021) to generate stable lasso selections.

[48] We include nonpsychometric questions into this prediction exercise and find that the first 13 variables chosen by the Lasso are all psychometric measures, as are the first 16 chosen by the Random Forest. This shows the predictive power of these measures while also clarifying that risk and cognitive measures still have important roles to play in understanding the heterogeneity in returns. Overall, we view each indicator alone as a poor proxy for what is a complicated set of interrelated characteristics, which we will later bundle under the term "optimism." It's worth noting that "optimism" in this case is an ex post interpretation of this bundle of characteristics by the authors of this paper and does not come directly from the psychology literature we reference above. We also include in online Appendix Table 6 all the other psychometric questions we asked at baseline and the differences between the top and bottom groups. The table shows that there are a lot of correlations between the measures and loan performance. These individual coefficients are, however, hard to interpret. We use the feature importance measures and Lasso as more systematic ways to identify the most robust and important correlates of performance.

Table 7 and online Appendix Table 8, where questions higher up the list are those that are ranked as more important by the method.

Both methods choose the following seven statements as the most important predictors of the individual treatment effect: (i) "I tend to act first and worry about consequences later," (ii) "I can think of several solutions to any problem," (iii) "When I make decisions I usually go with my first, gut feeling," (iv) "I always get things done ahead of time," (v) "I would work seven days a week if I could," (vi) "I spend a lot of time planning for my future," (vii) "I prefer to have a flexible schedule—I don't like being tied down." Across all of these questions, those who have high treatment effects are more likely to *disagree* with these statements. We consider these responses to be suggestive evidence that those who are in the bottom group are overly optimistic about their own ability to succeed. Motivated by this interpretation, the next section argues that our results (both the heterogeneity and reversal of fortune) are consistent with a simple model in which those who benefit most from the large loans are those who are careful and shrewd, while the poor performers are overly optimistic about themselves and act impetuously.[49]

Motivated by the suggestive evidence regarding optimism, we manually consider the remaining psychometric statements that we collected at baseline and identify four statements that we feel closely align with optimism. In particular, we look at responses to (i) "In life, failure is not an option," (ii) "When I make a business decision it is almost always the right decision," (iii) "I have always believed I am going to be successful," and (iv) "I prefer to focus on opportunities rather than risks." In all four cases, poor performers are more likely to agree with these statements and hence are more optimistic.

In summary, the data suggest that firms that performed well with big loans are fronted by entrepreneurs who have relatively higher cognitive capacity, are more risk averse, and are less optimistic/overconfident in their abilities. Standard facts about the firm itself are far less important in explaining relative performance.

VI. An Explanation from Optimism and Risk Taking

Our main findings, particularly the reversal of fortune documented in Table 5, may seem surprising. The result is inconsistent with many models used to study capital misallocation, which tend to assume that entrepreneurs draw a productivity type z_i, which multiplies a DRS production function to give output $z_i f(k_i)$ (e.g., Banerjee and Moll 2010). Motivated by the evidence above, which suggests that those who do poorly are perhaps overoptimistic or risk loving, this section shows that our results are in fact easily generated by combining two reasonable assumptions: (i) heterogeneous levels of optimism, with some individuals being overoptimistic, and (ii) decreasing marginal returns to risk taking. Because we think these assumptions are likely to hold broadly, we believe that our findings are likely to be relevant in a range of settings. While we refer throughout to optimism, we think of this as a set

[49] Those in the lowest treatment group, the overoptimists, indicate stronger agreement with several statements. Our interpretations are "I tend to act first … " and "When I make decisions …" and "I prefer to have a flexible schedule …" indicate impetuousness; "I can think of several solutions …," "I always get things done …," "I would work seven days a week …," "I spend a lot time planning …" indicate hard work but also a lack of forethought and realism.

of related characteristics. An optimist is someone who believes that things will work out well. This might be a general belief or a belief in oneself. The latter case is usually termed self-confidence, which we see as a subset of optimism. There is also a close relationship with risk tolerance. As is well known, it is empirically difficult to separate risk tolerance from optimism: someone who is optimistic believes that risks will turn out well and so is more likely to take risks. Because we cannot easily distinguish these characteristics using our data, we keep them bundled under the heading "optimism." It is important to note that our aim is to show that our results are *consistent* with a simple and realistic model, and more research would be required to test the specifics of the model.[50]

We first motivate the theoretical framework with a stylized example. Consider two risk-averse entrepreneurs who each own a restaurant. Omar is an optimist, and Rana is a realist, in a sense that will become clear. Both have done well with their business and wish to expand and apply for a loan to do so. If they are placed in the control group, they each have a constrained set of projects they can pursue. There is a low risk choice, replacing some kitchen equipment, which will lower costs and lead to a certain increase in profit, and a high risk choice, expanding into a neighboring building to increase the size of the business, which will only pay off if demand is high. At this level of investment, the high-risk option has a higher expected return. Rana is realistic about the probability that the expansion will work, and, given her risk aversion, chooses the low-risk option. Omar would choose the same if he were realistic, but his optimism leads to him overestimating the probability that the expansion will succeed—he is optimistic about market demand and/or overconfident about his own talents, and this optimism overcomes his risk aversion. In expectation, Omar sees a larger increase in profits than does Rana.

Next, suppose Omar and Rana are placed in the treatment group and receive a larger loan. The larger loan opens up new possibilities, and both entrepreneurs again have two different options: continue expanding their existing restaurant or open a second location on the other side of town. Because of decreasing returns to risk taking, the first, safer option now has a higher expected return than the second, riskier option. Rana, being the realist, realizes that overexpansion has a lower return and decides to continue growing her original restaurant. With the larger loan, she is able to take more action and sees a larger increase in her profits than with the small loan. Omar, on the other hand, incorrectly believes that he can handle expansion to a second location. He perceives the large expansion as having a higher expected return. In reality, the second-location expansion is too risky, a fact that Omar's optimism has caused him to ignore; the second location performs poorly, and the large loan causes a *decrease* in Omar's profits. As noted above, the same results could be obtained if Rana were risk averse and Omar were risk loving. In this case, Omar's convex utility function could lead him to take a decision that reduces his expected profits.

Omar and Rana's stories are consistent with our empirical findings: first, there is heterogeneity in the impacts of the large loan, and the large loan causes Omar's profits to decrease; he is a poor performer. Second, we see a reversal of fortune—the

[50] In online Appendix Table 4, we report the correlations between psychometric variables and our measures of risk tolerance and cognitive ability. We find that they are not highly correlated.

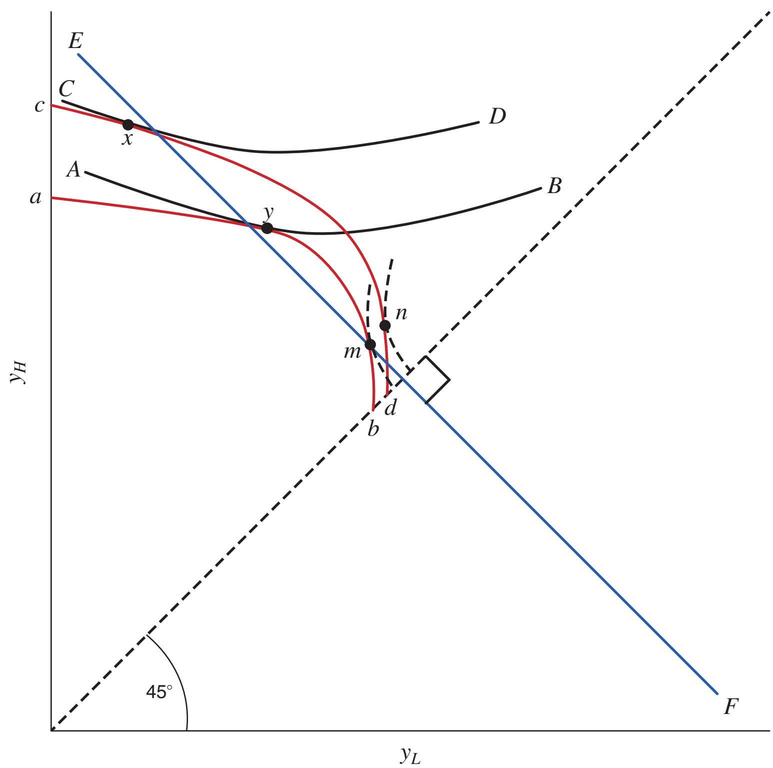

FIGURE 3. THE MODEL

optimist Omar does better with the small loan, and the realist Rana does better with the large loan.

Slightly more formally, assume that each entrepreneur in our experiment has a range of projects to choose from. Each project $p \in P$ yields income y_H^p in a "high" state of the world and y_L^p in a "low" state of the world. We assume that the true probability of state H is π_H. The set of projects in P leads to a production possibility frontier (PPF). Line *ab* in Figure 3 shows what we consider to be a reasonable PPF. It has two key characteristics. First, there is a range of projects over which an increase in risk is rewarded with an increase in expected return. In the figure, we assume that $\pi_H = \frac{1}{2}$ so that the line *EF*, which is perpendicular to the 45-degree line, marks an iso-expected return line. The PPF is steeper than this iso-expected return for the type of modest projects that Rana chose above. Second, and this is our first key assumption, the expected return to risk taking is decreasing in the amount of risk. It seems unlikely that the usual risk-return relationship should hold indefinitely; continuing to increase y_H, even at the cost of lowering y_L, is likely to be a difficult task. This argument suggests that at some point, y_H can only be increased further at the expense of reducing expected return. As in the story above, a large-scale expansion is very risky and may be accompanied by a reduction in expected returns. The concavity of line *ab* in Figure 3 captures these ideas, and the fact that it becomes less steep than *EF* implies that more risk taking eventually leads to a reduction in expected returns.

Next, it seems reasonable to assume that the set of available projects depends on loan size. We make two assumptions. First, any project that is feasible with a small loan is feasible with a large loan. This should be uncontroversial and implies that the PPF with a large loan lies everywhere above the PPF with a small loan. Second, the PPF with large loans has a greater (absolute) slope for every y_L. This amounts to assuming that larger loans allow for a greater return to risk taking. In terms of the example above, Omar could surely have expanded to a second location even with a small loan, but that risk would almost surely come with a low return. Overexpanded and strapped for cash, there is likely a very low reward for this extreme level of risk taking. With a larger loan, the situation is probably somewhat better, with each unit of expansion more likely to be rewarded with a positive return. Figure 3 again shows the implications. Curve *ab* represents a possible PPF for a small loan, while *cd* is a possible PPF for a large loan.

Finally, we add our second key assumption, that optimism is heterogeneous, with some individuals being overoptimistic. We capture this by assuming that there are two types of entrepreneurs: *O* for optimists and *R* for realists. We denote the beliefs of optimists π_H^O and the beliefs of realists π_H^R, and we assume $\pi_H^O > \pi_H^R$. That is, optimists believe that the outcome of any investment is more likely to be the good outcome. Second, we assume that optimists are "overoptimistic" in the sense that $\pi_H^O > \pi_H$. As noted above, this overoptimism could be because of a general belief that investments work out well or more of a self-belief or overconfidence. Both of these assumptions have strong support in the psychology literature (Carver, Scheier, and Segerstrom 2010; Peterson 2000; Weinstein and Klein 1996; Frese and Gielnik 2014; Hmieleski and Baron 2009; Hilary et al. 2016).[51] The two entrepreneurs solve

$$\max_{p \in P}\left(\pi_H^e u\left(y_H^p\right) + \left(1 - \pi_H^e\right) u\left(y_L^p\right)\right),$$

where $e \in \{O, R\}$ is the entrepreneur's type and the set P has the properties of the PPF described above.

We now wish to demonstrate two potential outcomes in our setting. First, our model can accommodate negative returns to a large loan for those who are optimists. Returning to Figure 3, the lines *AB* and *CD* are indifference curves for an optimist, with *CD* associated with higher utility. These indifference curves are less steep than the iso-expected return line *EF*, reflecting the optimists' belief that the high state is more likely than it actually is.[52] Given these indifference curves, point *x* is preferred to the point *y*. But point *x* has a lower expected return, which can be seen because points *x* and *y* lie on opposite sides of the line *EF*.[53] Just like Omar, the optimist here

[51] We could also model optimism as heterogeneity in the belief about the extent of decreasing returns to risk taking. This leads to precisely the same results as presented in the text.

[52] Recall that in this state space representation of the SEU model, indifference curves lie perpendicular to the fair odds line at the 45-degree line. In Figure 3 the line EF is the "true" fair odds line, which the dashed indifference curves of the realist are tangent to at the 45-degree line. The optimists' indifference curves are much flatter at the 45-degree line, indicating a belief that the good state of the world, *H*, is more likely to occur in the optimists' opinion.

[53] As noted above, a similar outcome could be achieved if we replaced the optimist with someone who is risk loving but cannot be achieved if all types of entrepreneurs are weakly risk averse because no risk-averse person would knowingly choose a project that reduces expected profits. We believe that the two explanations are very similar—they both imply a type of individual who, when given a chance to expand with a large loan, overexpands and loses money.

believes that a good outcome is more likely than it really is and pushes forward with a risky project that a more sober realist would avoid.

Second, our model can explain the reversal of fortune that we see in the experiment. Figure 3 again demonstrates. As noted, the curves *AB* and *CD* are indifference curves for the optimist. In contrast, the realist's indifference curves are the dashed lines parallel to the line *EF* on the 45-degree line. With the small loan PPF, the realist chooses point *m*, while the optimist chooses point *y*. The optimist does better with the small loan because his optimism causes him to take greater risk. Over some range greater risk, he is rewarded with greater expected return. With the larger loan, the optimist now chooses point *x*, which, as we noted above, reduces his expected earnings. In contrast, the realist chooses *n*, increasing her expected return and earnings more than the optimist with the large loan.

This simple model is consistent with our experimental findings. While it is hard to test the first key assumption—decreasing returns to risk taking—the previous section provided some evidence on the optimism or overconfidence assumption.[54]

VII. Policy Implications and Conclusion

Taken as a whole, our results suggest both that the within-lender allocation of credit matters for firm profitability and productivity and that standard approaches to determining how to allocate credit are likely inefficient. The results are striking, and—as with many such results—replication is warranted, ideally across markets and contexts, in order to understand whether the psychometric results are consistently predictive of heterogeneity. The underlying characteristics those data are picking up may interact with some contextual factors or loan features, and thus, replication and extensions could help understand these markets and entrepreneurs better.

Our work also suggests the potential efficacy of psychometric data in improving allocation. Caution is required before taking this step. Two important issues arise. First, our slate of psychometric questions are predictive of treatment effects but are also potentially subject to gaming. Individuals were informed that their loan decision would not be based on their answers to these questions, and so our results do not reveal what would happen if lending decisions depended on the responses. Finding strategy-proof ways to detect which individuals would benefit most from large loans is a fruitful avenue for future research. Second, characteristics such as age and gender, while themselves not strongly predictive of heterogeneity, are correlated with our

[54] We concentrate on providing evidence on heterogeneity in optimism, but we are able to provide some indirect evidence on decreasing returns to risk taking. As discussed in this section, poor performers take more risk when in the control group, which causes them to do better, but this same tendency to take more risk when in the treatment group leads them to reduce their profits. An indirect test of the decreasing returns hypothesis then would be to look only at poor performers and to test whether risk tolerance (which leads to higher risk taking) is correlated with higher profits when in control but lower profits when in treatment. To run these regressions, we regress profits on willingness to spend money on a risky investment, instrumented with self-reported willingness to take risks. These regressions reveal results that are broadly consistent with the theory. Among poor performers in the control group, profits increase about 1.8 percent for each additional percent they're willing to invest in a risky asset, while profits decrease 1.1 percent when they are in treatment. Unfortunately, these results are not well powered, with *p*-values of 0.48 and 0.45, respectively. Given the lack of power, and difficulty in giving a causal interpretation to these results, we have chosen not to emphasize them.

GATES groups, and using psychometric data to deny credit could lead to unintended discrimination against these groups.

We also find that many borrowers choose to take out a loan that is larger than what they are able to handle. We interpret this as evidence that some entrepreneurs are overly optimistic and hence more risk loving (as evidenced by their reported ability to "think of several solutions to any problem," and their greater reported risk-taking preferences). This highlights *belief elicitation* as an important gap in the literature on firm development. While some recent and important work has started tackling this (e.g., Hussam, Rigol, and Roth 2022; McKenzie 2018), much more needs to be learned both regarding how to elicit beliefs (both for researchers, i.e., from survey methods, and for practitioners, i.e., for targeting of services) as well as how to incorporate beliefs into our understanding of heterogeneity in returns to policies.[55,56]

Other limitations include the possibility that the treatment had spillover effects on other firms, which our research is not designed to uncover (e.g., Cai and Szeidl 2022). Additionally, our discussion of optimism is ex post, and it would be nice to see the results replicated in a study that prespecifies heterogeneity tests based on validated measures of optimism.

Our results show the important role credit expansion can have on firm growth, but strong caution is necessary since this heterogeneity can lead to both winners and losers. Many policymakers consider supporting firm growth to be a primary objective, but previous studies have shown that this is often ineffective on average for small loans. We show that a relatively large infusion of credit could help firms grow. Yet because of the heterogeneous treatment effects, this is not an economy-wide prescription. While some firms benefited from a large expansion in borrowing, others were harmed, even leading to firm exit. Making use of these findings therefore requires either a better understanding of how to improve the allocation of credit or a belief that expanding larger loans to all firms will let bad firms exit, opening up additional room for the most productive firms to flourish. While plausible, this second line of thinking requires careful consideration of the human costs associated with failed businesses.

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[55] We also find that responses to psychometric questions and beliefs can change over time, as evidenced by changes in responses when we ask the same question at baseline and again at follow-up. This implies that it's possible that who can benefit from a large loan can change over time, with the same person being able to benefit in one period but actually being a poor performer in another time period.

[56] This is also true of eliciting loan officer beliefs. While we explained to the loan officers that their answers would not affect lending decisions, they may have strategically misreported due to a misunderstanding in the process.

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